

Lahore University of Management Sciences

School of Science and Engineering

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Solution Scheme (Prepared by Dr. Hassan Mohy-ud-Din)

Course Title: **AI 500 – Foundations of AI**

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Type: **Quiz # 2**

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Marks: **20 points**

Duration: **30 minutes**

Predicting Customer Complaints for an Online Store [CLO 1, CLO 2]

An online retailer wants to develop an AI system that predicts whether a newly placed order is likely to result in a **customer complaint within 14 days**. The company hopes to use the predictions to identify orders that may need additional attention before shipment. The available dataset contains:

- product category;
- order value;
- delivery method;
- customer region;
- number of previous orders;
- number of previous complaints;
- customer age;
- website activity;
- payment method;
- final delivery time;
- whether the customer contacted support;
- whether a complaint was eventually filed.

During data preparation, the team discovers that:

- customer age is missing for 30% of customers because entering age is optional;
- some customers place several orders, so the same customer appears many times in the dataset;
- several orders appear twice because the payment system occasionally resubmits transactions;
- only 6% of orders result in complaints;
- the team replaces all missing ages with the mean calculated using the entire dataset before splitting it into training and test sets;
- individual orders are randomly divided between training and testing, so different orders from the same customer may occur in both sets;
- `final_delivery_time` and `whether_the_customer_contacted_support` are used as predictors even though both may become known only after the order has already been placed;
- the model achieves 94% accuracy.

Management concludes that the dataset has been cleaned successfully and that the model is ready for deployment.

(a) [2 + 2 points] Data Quality and Missing Values

i. Is a missing value for customer age necessarily a data-quality error? **Explain briefly**.

No. A missing age is not necessarily an error. In this case, age entry is optional, so a missing value may simply mean that the customer chose not to provide it.

Side Note: The important issue is therefore not merely whether a value is missing, but why it is missing and whether that missingness affects the intended analysis or prediction task. The absence itself may also carry information if certain types of customers are systematically more or less likely to provide their age.

Rubric

States that missing age is not automatically a data-quality error	1
Explains that optional entry/customer behavior or the reason for missingness must be considered	1

ii. Why might replacing every missing age with the overall mean be problematic? Give **two** reasons.

Students need two valid reasons. Possible answers include:

- It can distort the distribution. All missing observations receive exactly the same artificial age, producing an unrealistic concentration around the mean and reducing observed variance.
- It can distort relationships between variables. If missingness differs by customer group or behavior, replacing all missing ages with the same value can weaken or alter relationships between age and other variables.
- Missingness itself may be informative. A customer who chooses not to provide age may differ systematically from one who provides it. Mean imputation removes this information unless missingness is separately represented.
- In this case the mean was computed from the entire dataset. This allows information from the test set to influence preprocessing and therefore creates train-test contamination.

Rubric

Reason # 1	1
Reason # 2	1

(b) [2 + 1 points] Duplicates and Data Curation: The dataset contains duplicate order records caused by repeated transaction submission.

i. Why can duplicate records be harmful to analysis or machine learning?

Duplicate records may:

- inflate the apparent number of orders;
- distort frequencies and summary statistics;
- give duplicated transactions disproportionate influence during training;
- distort the apparent complaint rate;
- place the same underlying transaction in both training and testing;
- produce overly optimistic estimates of model performance.

Rubric

Identifies one valid consequence	1
Identifies/explains a second valid consequence	1

ii. What should the data team do before simply deleting them?

The team should first establish whether the records are true duplicates or legitimate separate orders. This could involve checking:

- order/transaction IDs;
- timestamps;

- payment identifiers;
- customer IDs;
- product/order details;
- source-system logs.

Confirmed duplicate submissions should then be resolved using a documented and reproducible deduplication rule, rather than deleting every repeated-looking record automatically.

Rubric

Clearly states that records must first be investigated/verified as duplicates before deletion.	1
Says "check the duplicates" without explaining what is being checked.	0.5

(c) [5 points] Data Leakage: Identify and explain **three distinct** leakage or contamination problems in the case. For each problem, identify what is wrong and state how it should be corrected. Your answer should consider preprocessing, repeated customers, and when information becomes available.

Problems	Brief Explanation
Preprocessing contamination	The mean age is calculated using the entire dataset before train-test splitting. First divide the data into training and test partitions. Calculate the mean using training data only, then apply that fixed training-derived value to both training and test data.
Group leakage	The same customer can have multiple orders, yet orders are randomly split. Consequently, one customer's orders may occur in both training and testing. Partition at the customer level so that $G_{\text{train}} \cap G_{\text{test}} = \emptyset$. All orders belonging to one customer must remain in the same partition.
Temporal/target leakage	The prediction is supposed to be made for a newly placed order, yet: - final_delivery_time cannot be known when the order is placed; - whether_the_customer_contacted_support may also only become known later. Define the prediction/index time precisely and retain only variables genuinely available by that time.

Rubric

Correctly identifies preprocessing contamination and appropriate correction	1.5
Correctly identifies group/customer leakage and appropriate correction	1.5
Correctly identifies temporal/target leakage and appropriate correction	1.5
Overall explanation clearly connects leakage to unrealistically optimistic evaluation	0.5
Total	5

(d) [2 + 2 points] Class Imbalance and Evaluation: Only 6% of orders result in a complaint, yet the model achieves 94% accuracy.

i. Explain why 94% accuracy may provide very little evidence that the model is useful.

Only 6% of orders result in complaints. Therefore 94% do not. A classifier that simply predicts no complaint for every single order would achieve an accuracy of 94% while detecting zero complaints. Therefore the reported accuracy does not demonstrate that the model has learned to identify the minority class at all.

Rubric

Recognizes severe class imbalance / dominance of non-complaints	1
Explicitly explains that always predicting "no complaint" already yields 94% accuracy yet detects no complaints	1

ii. Give **two** better ways of evaluating the model in this situation and briefly explain what each would reveal.

Pick any two from the following:

- Recall/sensitivity for complaints: Measures the proportion of actual complaint-producing orders correctly identified. Particularly important if missing a problematic order is costly.
- Precision: Among orders flagged as likely to generate complaints, measures how many actually do. Useful because excessive false alarms could overwhelm customer-service resources.
- F_1-score: Balances precision and recall and is useful when the positive class is rare.
- Precision-recall curve (AUPRC): Particularly informative for rare positive events because it focuses on performance for the minority class.
- Confusion matrix: Shows false positives, false negatives, true positives, and true negatives rather than hiding them inside a single accuracy value.
- Calibration If the model outputs complaint probabilities, determines whether predicted probabilities correspond to observed complaint frequencies.
- Subgroup analysis: Tests whether performance differs across customer regions, delivery methods, customer groups, etc.

Rubric

Measure/Analysis # 1 with explanation	1
Measure/Analysis # 1 without explanation	0.5
Measure/Analysis # 2 with explanation	1
Measure/Analysis # 2 without explanation	0.5

(e) [4 points] Fitness for Purpose and Governance: Before deployment, recommend **four improvements** to the dataset or development process. Your recommendations may address: (i) representativeness, (ii) missing-data handling, (iii) duplicates, (iv) customer-level partitioning, (v) leakage, (vi) class imbalance, (vii) documentation, (viii) privacy, (ix) provenance, **OR** (x) monitoring.

Pick any four of the list below:

Improvement	Brief Explanation
Handle missing age more carefully	optional age entry may itself reflect customer behavior.
Fit preprocessing only on training data	prevents train-test contamination and gives a valid estimate of generalization.
Deduplicate verified duplicate transactions	prevents inflated frequencies, artificial sample size, and evaluation contamination.
Split by customer	prevents group leakage and tests generalization to unseen customers.
Remove future information	prevents temporal leakage.
Address class imbalance appropriately	the complaint class represents only 6% of observations.
Use metrics appropriate for imbalance	accuracy hides minority-class failure.
Assess representativeness	a large dataset need not be representative; representativeness is about matching the deployment population and conditions.
Improve documentation	future users must understand how the dataset was constructed and under what circumstances it is fit for use.
Record provenance and lineage	enables reproducibility, debugging, auditing, and rollback.
Apply data minimization/privacy controls	collecting and retaining only information reasonably necessary for a legitimate purpose

Monitor the deployed data pipeline	data quality and model usefulness can change after deployment.

Rubric

4 × 0.5 marks for identifying the appropriate action;
4 × 0.5 marks for explaining why it is needed.